Redes:

Design of an intelligent orchestration plan for proactive slice management in 6G networks

Prova Oral

1.Referentes a fundamentos teóricos e conceituais do projeto temático em questão;
2.Referentes ao estado da arte na pesquisa no projeto temático em questão;
3.Referentes ao seu currículo e experiências profissionais de pesquisa;

Resumo

The evolution toward sixth-generation (6G) mobile networks entails a profound transformation of network infrastructure, characterized by openness, virtualization, disaggregation, and the distribution of network functions. These networks aim to deliver ultra-reliable low-latency communications (URLLC) for critical applications such as digital twins, extended reality (XR), haptic communications, and cyber-physical systems. To meet these requirements, the use of network slicing, OpenRAN, and edge computing becomes indispensable.

However, these innovations bring new management challenges: networks are now composed of distributed, heterogeneous, and dynamic functional units, often operated by different vendors. Effective management and adaptive orchestration of these resources, while considering Service Level Agreements (SLAs) and key performance indicators (KPIs) such as Quality of Service (QoS) and Quality of Experience (QoE), constitute a major challenge. The integration of distributed artificial intelligence mechanisms, autonomous decision-making, and multi-domain coordination is therefore essential to make these systems resilient, efficient, and service-oriented.

This project is situated within this demanding technical and scientific context. Its objective is to propose an innovative approach to intelligent network orchestration that is proactive, scalable, interoperable, and aligned with the requirements of open and mission-critical 6G networks.

Referências

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